Complete Tomato Agriculture & ICAR Research Guide

1. Overview and Botanical Information

The tomato (Solanum lycopersicum) is one of the most important vegetable crops grown globally. It belongs to the Solanaceae family. Originally from South America, it is now a staple in diets worldwide and a critical commercial crop. Tomatoes are cultivated for fresh market consumption as well as for processing into paste, ketchup, juice, and canned products. It is a day-neutral plant, which means its flowering is not strictly dependent on daylight hours, making it highly adaptable to various growing regions.

2. Soil Requirements and Preparation

Tomatoes exhibit significant versatility and can grow in soil types ranging from light sandy soils to heavy clay. However, the optimal soil for commercial tomato production is a well-drained sandy loam with a rich organic matter content. The preferred soil pH is between 6.0 and 7.0.

Soil Preparation:
The land should be ploughed deep (20-25 cm) to break any hardpans and allow for deep root penetration. This is followed by 2-3 harrowings to achieve a fine tilth. Prior to the final ploughing, well-rotted Farm Yard Manure (FYM) or compost at the rate of 20-25 tonnes per hectare should be incorporated into the soil. Soil solarization using transparent polyethylene sheets for 3-4 weeks prior to planting is highly recommended to manage soil-borne pathogens and weeds.

3. Climate and Weather Conditions

Tomatoes are warm-season crops and highly susceptible to frost. They require a frost-free period of at least 3-4 months. The optimal temperature for seed germination is 25-30 degrees C. For vegetative growth and fruit set, day temperatures of 25-28 degrees C and night temperatures of 15-20 degrees C are ideal.

Extreme Temperatures:
Temperatures exceeding 32 degrees C inhibit fruit set due to poor pollen viability, while temperatures below 10 degrees C delay physiological activities and result in poor color development. High humidity accompanied by continuous rainfall creates a favorable environment for fungal diseases such as Early and Late Blight. Therefore, dry climates with access to irrigation are considered most suitable for high-yield tomato production.

4. ICAR Contributions and Tomato Varieties

The Indian Council of Agricultural Research (ICAR), specifically through institutes like the Indian Institute of

Horticultural Research (IIHR) and the Indian Agricultural Research Institute (IARI), has made monumental contributions to tomato breeding.

Key ICAR Varieties & Hybrids:
- Arka Rakshak: A triple disease-resistant F1 hybrid (resistant to Tomato Leaf Curl Virus, Bacterial Wilt, and Early Blight) capable of yielding up to 100 tonnes per hectare. Known for firm fruits suitable for long-distance transport.
- Arka Samrat: Another high-yielding hybrid with triple disease resistance, producing large, firm, and deep red fruits.
- Pusa Ruby: A popular early-growing variety developed by IARI, known for its distinct tangy flavor and suitability for both fresh consumption and processing.
- Pusa 120: Resistant to root-knot nematodes.
- Arka Vikas: A high-yielding variety adapted to rainfed conditions and tolerant to moisture stress.

Recent ICAR research has heavily focused on developing climate-resilient varieties that can withstand heat stress and fluctuating water availability.

5. Agricultural Methodologies: Nursery and Transplanting

Nursery Management:
Tomatoes are primarily grown from seeds raised in nursery beds before transplanting. Raised beds of 1-meter width and convenient length are prepared. Seeds are treated with Trichoderma viride or Captan to prevent damping-off disease. Seedlings are typically ready for transplanting in 25-30 days when they reach a height of 10-15 cm and have 4-5 true leaves.

Transplanting and Spacing:
Transplanting is generally done in the late afternoon to minimize transplant shock. Standard spacing varies depending on the variety (determinate vs. indeterminate) but generally ranges from 60x45 cm to 90x60 cm. Staking is required for indeterminate varieties to keep fruits off the ground, reducing soil-borne infections and improving fruit quality.

6. Irrigation, Fertigation, and Remote Sensing

Irrigation:
Tomatoes require consistent moisture. Water stress during the flowering and fruit development stages leads to blossom drop, poor fruit set, and Blossom-End Rot. Drip irrigation is the most efficient method, reducing water usage by 40-50% and minimizing weed growth.

Fertigation & Remote Sensing:
Fertigation (the application of fertilizers through the irrigation system) is standard practice in precision agriculture, ensuring nutrients are delivered directly to the root zone. Remote sensing is increasingly used in commercial tomato farming. Drones equipped with multispectral cameras capture NDVI (Normalized Difference Vegetation Index) data to map canopy health, identify water-stressed zones, and optimize fertigation schedules in real-time.

7. Major Diseases, Pests, and IPM

Pests:
- Tomato Fruit Borer (Helicoverpa armigera): Bores into the fruit, causing severe yield loss. Managed by planting marigold as a trap crop and using pheromone traps.
- Whiteflies (Bemisia tabaci): Vectors for the devastating Tomato Leaf Curl Virus (ToLCV). Controlled using yellow sticky traps and systemic insecticides like Imidacloprid.

Diseases:
- Late Blight (Phytophthora infestans): Causes rapid browning and wilting of leaves and fruit rot. Spreads quickly in cool, wet weather. Managed with copper-based fungicides.
- Early Blight (Alternaria solani): Target-board-like concentric rings on older leaves. Managed through crop rotation and proper spacing for air circulation.
- Bacterial Wilt (Ralstonia solanacearum): Causes sudden wilting of the entire plant while still green. Extremely difficult to eradicate; management relies entirely on using resistant varieties like Arka Rakshak and crop rotation with non-solanaceous crops.

8. Harvesting and Post-Harvest Management

Tomatoes are harvested at different stages of maturity depending on the market:
- Mature Green Stage: For long-distance shipping.
- Breaker Stage: When color begins to change from green to tannish-yellow. Best for distant markets.
- Pink/Red Ripe Stage: For local markets and immediate consumption.
- Fully Ripe: For processing industries (ketchup, paste).

Post-harvest management involves careful sorting, grading based on size and color, and packing in ventilated corrugated boxes. Storing tomatoes at 10-15 degrees C with 85-90% relative humidity extends shelf life while maintaining quality.